

Chapter 20: Coordinate Systems in Space (Hệ Tọa độ Không gian)

Chapter Overview

Three-dimensional coordinate systems extend our understanding of geometry from the plane to space. The **Oxyz coordinate system** allows us to precisely locate points, describe vectors, and represent geometric objects like spheres, planes, and lines in three-dimensional space.

This chapter establishes the foundation for spatial geometry by introducing coordinates, distance formulas, and the equation of a sphere. These concepts are essential for understanding 3D graphics, engineering design, physics simulations, and navigation systems.

Real-world connections:

- GPS Navigation:** Locating positions on Earth using 3D coordinates
- Computer Graphics:** Rendering 3D models and animations
- Architecture:** Designing buildings and structures in 3D space
- Physics:** Describing motion and forces in three dimensions
- Astronomy:** Mapping positions of celestial bodies

20.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Coordinate system	/kou'ɔ:rdɪnət 'sɪstəm/	Hệ tọa độ	<i>The Oxyz coordinate system has three perpendicular axes.</i>
Three-dimensional space	/θri: di'menʃənəl speɪs/	Không gian ba chiều	<i>We live in three-dimensional space.</i>
Origin	/'ɔ:rdʒɪn/	Gốc tọa độ	<i>The origin O is the point (0, 0, 0).</i>

Axis (plural: axes)	<i>/ˈæksɪs/ (/ˈæksiːz/)</i>	Trục	<i>The three axes are Ox, Oy, and Oz.</i>
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Coordinate Elements

English	IPA	Vietnamese	Example
x-coordinate	<i>/eks kooˈɔːrdɪnət/</i>	Hoành độ	<i>The x-coordinate of point A is 3.</i>
y-coordinate	<i>/waɪ kooˈɔːrdɪnət/</i>	Tung độ	<i>The y-coordinate of point B is -2.</i>
z-coordinate	<i>/ziː kooˈɔːrdɪnət/</i>	Cao độ	<i>The z-coordinate represents height.</i>
Coordinate plane	<i>/kooˈɔːrdɪnət pleɪn/</i>	Mặt phẳng tọa độ	<i>The xy-plane has equation z = 0</i>

Geometric Objects

English	IPA	Vietnamese	Example
Point	<i>/pɔɪnt/</i>	Điểm	<i>Point M has coordinates (2, -1, 3).</i>
Vector	<i>/ˈvektər/</i>	Vector	<i>Vector AB connects points A and B.</i>
Sphere	<i>/sfɪr/</i>	Mặt cầu	<i>A sphere is the set of points equidistant from a center.</i>
Radius	<i>/ˈreɪdiəs/</i>	Bán kính	<i>The sphere has radius 5.</i>

Center	<i>/'sentər/</i>	Tâm	<i>The center of the sphere is at (1, 2, -3).</i>
Diameter	<i>/daɪ'æmɪtər/</i>	Đường kính	<i>The diameter equals twice the radius.</i>

Distance and Measurement

English	IPA	Vietnamese	Example
Distance	<i>/'dɪstəns/</i>	Khoảng cách	<i>The distance between two points is calculated using a formula.</i>
Distance formula	<i>/'dɪstəns 'fɔ:rmjələ/</i>	Công thức khoảng cách	<i>The distance formula extends the Pythagorean theorem to 3D.</i>
Midpoint	<i>/'mɪdpɔɪnt/</i>	Trung điểm	<i>The midpoint divides a segment into two equal parts.</i>

20.2 The Oxyz Coordinate System

Definition and Structure

Oxyz Coordinate System

The **Oxyz coordinate system** (or **3D Cartesian coordinate system**) consists of:

- An **origin** **O**
- Three mutually perpendicular axes: **Ox** (x-axis), **Oy** (y-axis), **Oz** (z-axis)
- Each axis has a positive direction indicated by an arrow

The three axes divide space into **eight octants** (similar to four quadrants in 2D).

Convention: We typically use a **right-handed coordinate system** where:

- The x-axis points "right"
- The y-axis points "forward" or "into the page"
- The z-axis points "up"

Coordinate Planes

The three axes determine three **coordinate planes**:

1. **xy-plane** (Oxy): Contains the x-axis and y-axis, equation
 $z = 0$
 2. **xz-plane** (Oxz): Contains the x-axis and z-axis, equation
 $y = 0$
 3. **yz-plane** (Oyz): Contains the y-axis and z-axis, equation
 $x = 0$
-

20.3 Coordinates of Points in Space

Point Coordinates

Point in Space

Every point

M

in space can be uniquely represented by an ordered triple

(x, y, z)

called its **coordinates**:

- x
= x-coordinate (hoành độ)
- y
= y-coordinate (tung độ)
- z
= z-coordinate (cao độ)

We write

$$M(x, y, z)$$

.

Geometric interpretation: To locate point

$$M(x, y, z)$$

:

1. Move

x

units along the x-axis

2. Move

y

units parallel to the y-axis

3. Move

z

units parallel to the z-axis

Special Points

- **Origin:**

$$O(0, 0, 0)$$

- **On x-axis:**

$$(x, 0, 0)$$

- **On y-axis:**

$$(0, y, 0)$$

- **On z-axis:**

$$(0, 0, z)$$

- **On xy-plane:**

$$(x, y, 0)$$

- **On xz-plane:**

$$(x, 0, z)$$

- On yz-plane:
 $(0, y, z)$
-

20.4 Vectors in Space

Vector Coordinates

Vector in Space

A vector

v

in space can be represented by coordinates

(a, b, c)

:

$$v = (a, b, c)$$

where

a

,

b

,

c

are the components along the x, y, z axes respectively.

Vector from Two Points

Vector Between Points

Given two points

$A(x_1, y_1, z_1)$

and

$B(x_2, y_2, z_2)$

, the vector from A to B is:

$$AB = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

Vector Operations

1. Addition:

$$u + v = (a_1 + a_2, b_1 + b_2, c_1 + c_2)$$

2. Scalar multiplication:

$$kv = (ka, kb, kc)$$

3. Dot product:

$$u \cdot v = a_1 a_2 + b_1 b_2 + c_1 c_2$$

4. Magnitude:

$$|v| = \sqrt{a^2 + b^2 + c^2}$$

20.5 Distance Formula in Space

Distance Between Two Points

Distance Formula

The distance between two points

$$A(x_1, y_1, z_1)$$

and

$$B(x_2, y_2, z_2)$$

is:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

This is the **3D extension of the Pythagorean theorem**.

Derivation: The distance formula comes from applying the Pythagorean theorem twice:

1. First in the xy-plane to find the horizontal distance
2. Then in the vertical direction to include the z-component

Midpoint Formula

Midpoint Formula

The midpoint M of the segment connecting

$$A(x_1, y_1, z_1)$$

and

$$B(x_2, y_2, z_2)$$

is:

$$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

20.6 Equation of a Sphere

Definition

Sphere

A **sphere** with center

$$I(a, b, c)$$

and radius

$$R > 0$$

is the set of all points

$$M(x, y, z)$$

in space such that:

$$IM = R$$

Standard Equation

Equation of a Sphere

A sphere with center

$$I(a, b, c)$$

and radius

$$R$$

has the equation:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

Special case: Sphere centered at the origin:

$$x^2 + y^2 + z^2 = R^2$$

General Form

A sphere can also be written in **general form**:

$$x^2 + y^2 + z^2 + Dx + Ey + Fz + G = 0$$

This represents a sphere if and only if:

$$D^2 + E^2 + F^2 - 4G > 0$$

In this case:

- Center:

$$I\left(-\frac{D}{2}, -\frac{E}{2}, -\frac{F}{2}\right)$$

- Radius:

$$R = \frac{1}{2}\sqrt{D^2 + E^2 + F^2 - 4G}$$

Position of a Point Relative to a Sphere

For a sphere

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

and point

$$M(x_0, y_0, z_0)$$

:

- M

is **on the sphere** if

$$(x_0 - a)^2 + (y_0 - b)^2 + (z_0 - c)^2 = R^2$$

- M

is **inside the sphere** if

$$(x_0 - a)^2 + (y_0 - b)^2 + (z_0 - c)^2 < R^2$$

• M

is **outside the sphere** if

$$(x_0 - a)^2 + (y_0 - b)^2 + (z_0 - c)^2 > R^2$$

20.7 Worked Examples

Example 1: Finding Distance Between Points

Problem: Find the distance between points

$$A(1, -2, 3)$$

and

$$B(4, 2, -1)$$

.

Solution:

Using the distance formula:

$$AB = \sqrt{(4 - 1)^2 + (2 - (-2))^2 + (-1 - 3)^2}$$

$$= \sqrt{3^2 + 4^2 + (-4)^2}$$

$$= \sqrt{9 + 16 + 16}$$

$$= \sqrt{41}$$

Answer:

$$AB = \sqrt{41} \approx 6.40$$

units

Example 2: Finding Midpoint

Problem: Find the midpoint of the segment connecting

$$A(2, -1, 5)$$

and

$$B(6, 3, -3)$$

.

Solution:

Using the midpoint formula:

$$M = \left(\frac{2+6}{2}, \frac{-1+3}{2}, \frac{5+(-3)}{2} \right)$$

$$= \left(\frac{8}{2}, \frac{2}{2}, \frac{2}{2} \right)$$

$$= (4, 1, 1)$$

Answer: The midpoint is

$$M(4, 1, 1)$$

Example 3: Finding Vector Coordinates

Problem: Given points

$$A(1, 2, -3)$$

and

$$B(4, -1, 2)$$

, find:

a) The coordinates of vector

$$\vec{AB}$$

b) The magnitude of

$$\vec{AB}$$

Solution:

a)

$$\vec{AB} = (4 - 1, -1 - 2, 2 - (-3)) = (3, -3, 5)$$

b)

$$|\vec{AB}| = \sqrt{3^2 + (-3)^2 + 5^2} = \sqrt{9 + 9 + 25} = \sqrt{43}$$

Answer: a)

$$\vec{AB} = (3, -3, 5)$$

; b)

$$|AB| = \sqrt{43} \approx 6.56$$

Example 4: Equation of a Sphere (Given Center and Radius)

Problem: Write the equation of a sphere with center

$$I(2, -3, 1)$$

and radius

$$R = 4$$

.

Solution:

Using the standard form:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

Substituting

$$a = 2$$

,

$$b = -3$$

,

$$c = 1$$

,

$$R = 4$$

:

$$(x - 2)^2 + (y + 3)^2 + (z - 1)^2 = 16$$

Answer:

$$(x - 2)^2 + (y + 3)^2 + (z - 1)^2 = 16$$

Example 5: Finding Center and Radius from Equation

Problem: Given the sphere

$$(x + 1)^2 + y^2 + (z - 2)^2 = 9$$

, find:

a) The center

b) The radius

c) Whether point

$$P(2, 0, 2)$$

is inside, on, or outside the sphere

Solution:

a) Rewrite as

$$(x - (-1))^2 + (y - 0)^2 + (z - 2)^2 = 3^2$$

Center:

$$I(-1, 0, 2)$$

b) Radius:

$$R = 3$$

c) Calculate

$$IP^2$$

:

$$IP^2 = (2 - (-1))^2 + (0 - 0)^2 + (2 - 2)^2 = 3^2 + 0 + 0 = 9$$

Since

$$IP^2 = R^2 = 9$$

, point P is **on the sphere**.

Answer: a) Center

$$I(-1, 0, 2)$$

; b) Radius

$$R = 3$$

; c) P is on the sphere

Example 6: Sphere with Diameter Endpoints

Problem: Find the equation of the sphere with diameter AB, where

$$A(1, 2, 1)$$

and

$$B(3, 4, 5)$$

.

Solution:

The center is the midpoint of AB:

$$I = \left(\frac{1+3}{2}, \frac{2+4}{2}, \frac{1+5}{2} \right) = (2, 3, 3)$$

The radius is half the length of AB:

$$AB = \sqrt{(3-1)^2 + (4-2)^2 + (5-1)^2} = \sqrt{4 + 4 + 16} = \sqrt{24} = 2\sqrt{6}$$

$$R = \frac{AB}{2} = \sqrt{6}$$

The equation is:

$$(x-2)^2 + (y-3)^2 + (z-3)^2 = 6$$

Answer:

$$(x-2)^2 + (y-3)^2 + (z-3)^2 = 6$$

Example 7: Converting General Form to Standard Form

Problem: Show that

$$x^2 + y^2 + z^2 - 4x + 6y - 2z + 5 = 0$$

represents a sphere, and find its center and radius.

Solution:

Complete the square for each variable:

$$x^2 - 4x + y^2 + 6y + z^2 - 2z + 5 = 0$$

$$(x^2 - 4x + 4) + (y^2 + 6y + 9) + (z^2 - 2z + 1) = -5 + 4 + 9 + 1$$

$$(x-2)^2 + (y+3)^2 + (z-1)^2 = 9$$

This is a sphere with center

$$I(2, -3, 1)$$

and radius

$$R = 3$$

.

Answer: Center

$$(2, -3, 1)$$

, radius

$$R = 3$$

Example 8: Determining if an Equation Represents a Sphere

Problem: Determine whether each equation represents a sphere:

a)

$$x^2 + y^2 + z^2 - 2x + 4y - 6z + 20 = 0$$

b)

$$x^2 + y^2 + z^2 + 4x - 2y + 2z - 3 = 0$$

Solution:

For the general form

$$x^2 + y^2 + z^2 + Dx + Ey + Fz + G = 0$$

to represent a sphere:

$$D^2 + E^2 + F^2 - 4G > 0$$

a)

$$D = -2$$

,

$$E = 4$$

,

$$F = -6$$

,

$$G = 20$$

$$(-2)^2 + 4^2 + (-6)^2 - 4(20) = 4 + 16 + 36 - 80 = -24 < 0$$

This does **not** represent a sphere.

b)

$$D = 4$$

,

$$E = -2$$

,

$$F = 2$$

,

$$G = -3$$

$$4^2 + (-2)^2 + 2^2 - 4(-3) = 16 + 4 + 4 + 12 = 36 > 0$$

This **does** represent a sphere.

Center:

$$I(-2, 1, -1)$$

, Radius:

$$R = \frac{1}{2}\sqrt{36} = 3$$

Answer: a) Not a sphere; b) Sphere with center
 $(-2, 1, -1)$
 and radius 3

20.8 Applications

Application 1: GPS and Earth Coordinates

Problem: In a simplified model, Earth is a sphere with radius 6371 km. Using a coordinate system with center at Earth's center, a satellite is at position

$$S(4000, 3000, 5000)$$

(in km). How far is the satellite from Earth's surface?

Solution:

The distance from the satellite to Earth's center is:

$$OS = \sqrt{4000^2 + 3000^2 + 5000^2} = \sqrt{16000000 + 9000000 + 25000000}$$

$$= \sqrt{50000000} = 1000\sqrt{50} \approx 7071.07 \text{ km}$$

The distance from the satellite to Earth's surface is:

$$7071.07 - 6371 = 700.07 \text{ km}$$

Answer: The satellite is approximately 700 km above Earth's surface.

Application 2: 3D Design and Architecture

Problem: An architect designs a spherical dome with center at

$(0, 0, 10)$

meters and radius 10 meters. The ground is the xy -plane (

$z = 0$

). Find the radius of the circular base where the dome meets the ground.

Solution:

The sphere equation is:

$$x^2 + y^2 + (z - 10)^2 = 100$$

At ground level (

$z = 0$

):

$$x^2 + y^2 + 100 = 100$$

$$x^2 + y^2 = 0$$

Wait, this gives only the point

$(0, 0, 0)$

. Let me reconsider...

Actually, if the dome has center at

$(0, 0, 10)$

and radius 10, then the lowest point is at

$z = 0$

. At

$z = 0$

:

$$x^2 + y^2 + (0 - 10)^2 = 100$$

$$x^2 + y^2 + 100 = 100$$

$$x^2 + y^2 = 0$$

This means the dome just touches the ground at one point. For a dome with a circular base, the center should be below ground. Let's say the center is at

$$(0, 0, 5)$$

and radius is 10:

$$x^2 + y^2 + (z - 5)^2 = 100$$

At

$$z = 0$$

:

$$x^2 + y^2 + 25 = 100$$

$$x^2 + y^2 = 75$$

The base circle has radius

$$r = \sqrt{75} = 5\sqrt{3} \approx 8.66$$

meters.

Answer: The circular base has radius

$$5\sqrt{3} \approx 8.66$$

meters.

Application 3: Collision Detection in 3D Games

Problem: In a 3D game, a player at position

$$P(2, 3, 1)$$

has a detection radius of 5 units. An enemy is at position

$$E(5, 7, 4)$$

. Is the enemy within detection range?

Solution:

Calculate the distance PE:

$$PE = \sqrt{(5 - 2)^2 + (7 - 3)^2 + (4 - 1)^2}$$

$$= \sqrt{9 + 16 + 9} = \sqrt{34} \approx 5.83$$

Since

$$5.83 > 5$$

, the enemy is **outside** the detection range.

Answer: No, the enemy is not within detection range (distance ≈ 5.83 units).

20.9 Practice Exercises

Exercise 1: Find the distance between points

$$A(3, -1, 2)$$

and

$$B(-1, 2, 5)$$

.

Exercise 2: Find the midpoint of the segment connecting

$$P(1, 4, -2)$$

and

$$Q(5, -2, 6)$$

.

Exercise 3: Given

$$A(2, 1, -3)$$

and

$$B(5, -2, 1)$$

, find the coordinates and magnitude of vector

$$\overrightarrow{AB}$$

.

Exercise 4: Write the equation of a sphere with center

$$(3, -1, 4)$$

and radius 6.

Exercise 5: Find the center and radius of the sphere

$$x^2 + y^2 + z^2 - 6x + 4y - 2z + 5 = 0$$

.

20.10 Chapter Summary

Key Concepts

1. Oxyz Coordinate System

- Three mutually perpendicular axes: Ox, Oy, Oz
- Origin O at $(0, 0, 0)$
- Three coordinate planes: xy, xz, yz

2. Point Coordinates

- Every point M has unique coordinates (x, y, z)
- Special points on axes and planes

3. Vectors in Space

- Vector $v = (a, b, c)$
- $AB = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$
- Magnitude:
 $|v| = \sqrt{a^2 + b^2 + c^2}$

4. Distance Formula

- $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
- Midpoint:
 $M(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2})$

5. Sphere Equation

- Standard form:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

- Center:

$$I(a, b, c)$$

, Radius:

$$R$$

- General form:

$$x^2 + y^2 + z^2 + Dx + Ey + Fz + G = 0$$

20.11 Review Exercises

Exercise 1: In the Oxyz coordinate system, plot the following points and describe their positions:

a)

$$A(3, 0, 0)$$

b)

$$B(0, -2, 0)$$

c)

$$C(0, 0, 4)$$

d)

$$D(2, 3, -1)$$

Exercise 2: Given points

$$A(1, 2, 3)$$

,

$$B(4, 0, -1)$$

, and

$$C(-2, 3, 5)$$

:

a) Find

$$AB$$

,

$$BC$$

, and

$$C(-2, 3, 5)$$

:

a) Find

$$AB$$

,

$$BC$$

, and

$$AC$$

b) Determine if triangle ABC is equilateral, isosceles, or scalene

Exercise 3: Find the equation of the sphere:

a) Center at

$$(1, -2, 3)$$

, radius 5

b) Center at origin, passing through

$$(3, 4, 0)$$

c) Diameter with endpoints

$$A(2, 1, -1)$$

and

$$B(4, 3, 5)$$

Exercise 4: Determine whether the following equations represent spheres. If yes, find the center and radius:

a)

$$x^2 + y^2 + z^2 - 4x + 6y - 8z + 4 = 0$$

b)

$$x^2 + y^2 + z^2 + 2x - 4y + 6z + 20 = 0$$

Exercise 5: A weather balloon is released from point

$$A(0, 0, 0)$$

and travels to point

$$B(300, 400, 500)$$

(coordinates in meters). How far did the balloon travel?

20.12 Memory Aids

Distance Formula Mnemonic

"3D Pythagoras: Square the differences, add them up, take the root"

$$\text{Distance} = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

Sphere Equation Pattern

"Center shifts, radius squares"

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

- Center
 (a, b, c)
appears with **minus signs**
- Radius
 R
is **squared**

Midpoint Formula

"Average the coordinates"

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Vector from Two Points

"End minus Start"

$$\vec{AB} = B - A = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

20.13 Common Mistakes to Avoid

 **Mistake 1:** Forgetting the square root in the distance formula

- **Wrong:**
 $AB = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$

- **Correct:**

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

⚠ **Mistake 2:** Sign errors in sphere equations

- **Wrong:**

$$(x + 2)^2 + (y - 3)^2 + (z + 1)^2 = 16$$

has center

$$(2, -3, 1)$$

- **Correct:** Center is

$$(-2, 3, -1)$$

(opposite signs)

⚠ **Mistake 3:** Forgetting to square the radius

- **Wrong:**

$$(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 5$$

has radius 5

- **Correct:** Radius is

$$\sqrt{5}$$

, not 5

⚠ **Mistake 4:** Incorrect vector direction

- $AB = B - A$

, not

$$A - B$$

- Order matters:

$$AB = -BA$$

⚠ **Mistake 5:** Not checking if general form represents a sphere

- Must verify

$$D^2 + E^2 + F^2 - 4G > 0$$

- If this is negative or zero, the equation does not represent a sphere
-

 **Congratulations!** You have completed Chapter 20: Coordinate Systems in Space. You now understand how to work with 3D coordinates, calculate distances, represent vectors, and write equations of spheres—essential skills for spatial geometry and real-world applications!